A Review Of Multi-Label Class Imbalance Learning

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(Survey Paper)

Abstract—**Multi-label learning allows each instance to be associated with multiple class labels simultaneously and has been widely used in fields such as medical diagnosis, text classification, and image recognition. However, class imbalance is common in multi-label data and can significantly affect model performance and generalization. In multi-label learning, this problem is usually reflected in three forms: intra-label imbalance, inter-label imbalance, and label-subset imbalance. To address this issue, this paper presents a systematic review of multi-label class imbalance learning. The formal definitions, theoretical challenges, and dataset characterization methods are introduced first. Representative studies are then reviewed from two perspectives: machine learning methods and deep learning methods. The former mainly includes sampling, cost-sensitive learning, threshold moving, other single methods, and ensemble learning, while the latter is further discussed under moderate and extreme imbalance settings. In addition, commonly used evaluation metrics, public datasets, software tools, and typical applications are summarized. The development trends and future research directions of multi-label class imbalance learning are also discussed. This review aims to serve as a valuable reference for researchers in this field by summarizing existing methods, offering critical analysis, and providing practical guidance.**

Index Terms—**Imbalanced data, multi-label classification, class imbalance learning, machine learning, imbalanced classification approaches, review on imbalanced classification.**

I. Introduction

In the field of machine learning, traditional single-label learning is the most widely studied paradigm in supervised learning, where each instance is associated with only one unique label. Over the past few decades, single-label learning methods have gradually matured and achieved significant results [1], [2], [3]. However, with the increasing complexity of application scenarios, the limitations of single-label learning are becoming increasingly apparent. Specifically, many phenomena in the real world are usually associated with multiple elements simultaneously. For instance, a movie might simultaneously encompass multiple genres, as shown in Fig. 1; a weather change might be accompanied by various phenomena such as rainfall, lightning, strong winds, and temperature drops; a protein might simultaneously exhibit multiple biological properties and functions. It is clear that these multidimensional and complex real-world problems pose significant challenges. Therefore, in recent years, a new supervised learning paradigm, multi-label learning, has received extensive attention from researchers [4]. Multi-label learning allows the assignment of multiple distinct class labels to a single instance, which better reflects the data information and underlying structure. It has a wide range of applications in key areas such as image recognition [5], medical diagnosis [6], recommendation systems [7], [8], bioinformatics [9], and optimization algorithms [10], [11].

Fig. 1. An example of a multi-label instance.

In single-label learning, class imbalance learning has long been a research hotspot, with rich research results due to its widespread existence and significant impact on model performance [12]. However, these methods are difficult to apply directly to multi-label learning to solve the multi-label class imbalance learning (ML-CIL) problem because instances of multi-label learning are simultaneously associated with multiple labels, each of which may exhibit different distributions [13], and the same instance may belong to both the majority and minority classes for different labels. In addition to this intuitive intra-label imbalance, there is also an imbalance problem among labels, and among subsets of labels [14]. Compared to the single-label class imbalance learning problem, the ML-CIL problem significantly increases the difficulty of model fitting and severely affects the prediction performance for minority labels. Therefore, it can be said that ML-CIL is an important challenge for multi-label learning and has attracted the attention of many researchers.

Although ML-CIL algorithms have become increasingly diverse, systematic reviews are still relatively scarce. To the best of our knowledge, the only review to date is by Tarekegn et al. [14], which initially summarized the ML-CIL field. However, with the rapid progress in recent years, the aforementioned

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TABLE I
NOTATION SYSTEM USED IN THIS REVIEW

Symbols	Meaning
X=R^d	d-dimensional instance space
$Y=\{l_j \| j=1,2,...,q\}$	set of all q labels, where l_j denotes the j-th label
$D=\{(x_i,y_i) \| i=1,2,\ldots,m\}$	multi-label training set with m instances
$S=\{(x_i,y_i) \| i=1,2,\ldots,p\}$	multi-label test set with p instances
$x_i=(x_{i1},x_{i2},...,x_{id})^{\mathrm{T}}$	returns d-dimensional feature vector of the i-th instance
$y_i \subseteq Y$	returns the actual label set for x_i
$y_{ij} \in \{0,1\}$	indicates whether label l_j is relevant to x_i, where y_{ij}=1 if $l_j \in y_i$, and 0 otherwise
$\hat{y}_i \subseteq Y$	returns the predicted label set for x_i
$\hat{y}_{ij} \in \{0,1\}$	indicates whether label l_j is predicted for instance x_i, where $\hat{y}_{ij}$=1 if $l_j \in \hat{y}_{ij}$, and 0 otherwise
$\hat{y}_i^*$	returns the predicted confidence scores for all labels of instance x_i, where $\hat{y}_{ij}^*$ indicates the confidence of label l_j for instance x_i
$\hat{y}_i^{\mathrm{Rank}}$	returns the ranking of labels for x_i, based on the descending order of $\hat{y}_i^*$, where $\hat{y}_{ij}^{\mathrm{Rank}}$ indicates the rank of label l_j for instance x_i
$\bar{y}_i$	returns the complement set of y_i
$y^\# \subseteq y_i$	label subset, where $F_{y^\#}$ indicates the frequency of label subset $y^\#$
$I(\cdot)$	indicator function, where I(true) = 1 and I(false) = 0
R_j	returns the instances associated with the label l_j
$\bar{R}_j$	returns the instances not associated with the label l_j
R_{Max}	returns the instance associated with the most frequent label
N_i^k	returns k nearest neighbors of the instance x_i
i, u	index of instances
j, o	index of labels
k	index of neighbors

review has become limited in both scope and depth. In particular, many recently proposed ML-CIL algorithms are not covered, and developments in deep learning–based ML-CIL have not been sufficiently discussed. Moreover, application scenarios and research trends have evolved significantly. Therefore, a new review is needed to provide a more comprehensive and systematic understanding of recent advances, clarify current challenges, and offer guidance for future research and applications. Compared with the previous review [14], this paper makes the following contributions:

1) We provide the most extensive review to date, covering 70 core algorithms—a 180% increase over prior work [14]—and present the first systematic survey of ML-CIL in deep learning.
2) We propose a novel and rigorous multi-level taxonomy, where technical routes are not only categorized into sampling, cost-sensitive learning, threshold-moving, other single methods, and ensemble learning, but also further decomposed into subcategories to reveal the underlying algorithmic mechanisms.
3) We present a comprehensive analysis and practical guide covering algorithms, evaluation metrics, datasets, and tools, while surveying emerging trends in the application of ML-CIL in fine-grained domains.

The remainder of this review is organized as follows: Section II introduces the background of ML-CIL. Section III presents the characterization metrics of multi-label datasets, while Section IV reviews machine learning-based solutions across various technical paradigms. Section V summarizes the developments of ML-CIL in the deep learning domain, and Section VI systematically organizes the evaluation metrics for ML-CIL models. Section VII summarizes commonly used datasets and software tools, whereas Section VIII discusses practical applications and future research directions. Finally, Section IX concludes this review, and the Appendix outlines the search strategy and statistical analysis of research trends.

II. Background

This section formally defines ML-CIL, outlines its theoretical challenges, and presents its taxonomy. Table I lists the symbols used in this review, whose definitions are not repeated in later sections.

A. Definition of Multi-Label Class Imbalance Learning

Class imbalance is one of the most challenging problems in classification tasks [12], [14], [15]. Unlike single-label class imbalance learning, which primarily involves significant differences in the number of instances across classes, ML-CIL is more complex because of its larger label space and more intricate label relationships.

We first revisit multi-label learning [4], [16]. Let X denote a d-dimensional feature space, and let Y denote a q-dimensional label space. Then, a multi-label training set with m instances can be represented as $D = \{(x_i,y_i)|\, i = 1,2,\ldots,m\}$, where $x_i \in X$ denotes a specific d-dimensional feature vector, and $y_i \subseteq Y$ represents the label set corresponding to instance x_i. The goal of a multi-label learning model is to construct a mapping function h that can accurately predict labels. For any unseen instance $x_i \in S$, the function $h(x)$ is expected to provide a predicted label set, i.e., $h(x_i) \to \hat{y}_i$.

However, in real-world multi-label data, the distribution of labels is often highly imbalanced, which makes multi-label class imbalance learning an important research paradigm. In this setting, the learning of the classifier h is influenced by several class imbalance factors simultaneously, which can generally be categorized into three types: intra-label imbalance, inter-label imbalance, and label-subset imbalance [14]. Let R_j denote the set of instances associated with label l_j, $\bar{R}_j$ denote the set of instances not associated with label l_j, and $F_{y^\#}$ denote the frequency of occurrence of label subsets $y^\#$. Assuming positive instances are always the minority, these imbalance types are defined as follows.

Definition 1. Intra-label imbalance: refers to the disparity between the numbers of positive and negative instances within a single label. Formally,

$$\exists j \in \{1, 2, \ldots, q\}, \ |R_j| \ll |\bar{R}_j| \tag{1}$$

This imbalance hinders the classifier's ability to accurately identify minority instances.

Definition 2. Inter-label imbalance: refers to differences in the numbers of positive instances across labels [17], [18], [19]. Formally,

$$\exists j, o \in \{1, 2, \ldots, q\}, \ |R_j| \ll |R_o| \tag{2}$$

This imbalance may bias the classifier toward majority labels.

Definition 3. ***Label-subset imbalance*** refers to the uneven distribution of label-subset occurrence frequencies in multi-label data. Specifically, some label subsets appear frequently, whereas others are rare or even absent [20], [21], [22]. Formally,

$$\exists y_a^{\#}, y_b^{\#} \subseteq Y, \ F_{y_a^{\#}} \ll F_{y_b^{\#}} \tag{3}$$

This may cause methods that consider label correlations [19], [23], [24] to be biased toward frequent label subsets.

B. Theoretical Challenges in Multi-Label Imbalance Learning

From a theoretical perspective, ML-CIL can be formulated based on the structural risk minimization (SRM) principle to balance empirical risk and model complexity. Accordingly, the learning objective is to optimize the model parameters θ by minimizing the following objective function:

$$\min_\theta \mathcal{L}\ (\theta) = \frac{1}{m} \sum_{i\,=\,1}^{m} \sum_{j\,=\,1}^{q} \updownarrow (y_{ij}, \hat{y}_{ij}(\theta)) + \lambda\Omega(\theta) \tag{4}$$

where $\hat{y}_{ij}(\theta)$ denotes the predicted value of label l_j for instance x_i, produced by the model parameterized by θ; $\updownarrow(\cdot)$ is the loss function; $\Omega(\theta)$ is a regularization term; and λ is a trade-off parameter. This formulation unifies existing ML-CIL methods, where different strategies can be interpreted as modifying the empirical loss term or implicitly reshaping the hypothesis space.

Therefore, the systematic bias introduced by multi-label class imbalance in empirical risk can also be explained from a theoretical perspective. For a specific label l_j, the gradient of the empirical loss with respect to parameters θ is derived as $\nabla_\theta \mathcal{L}_j\ (\theta) = \frac{1}{m}\ (\sum_{x_i \in R_j} \nabla_\theta \updownarrow(y_{ij}, \hat{y}_{ij}) + \sum_{x_i \in \bar{R}_j} \nabla_\theta \updownarrow(y_{ij}, \hat{y}_{ij})$. Under severe imbalance ($|\bar{R}_j| \gg |R_j|$), the gradient magnitude is largely dominated by negative samples, i.e., $||\sum_{x_i \in \bar{R}_j} \nabla_\theta \updownarrow(\cdot)|| \gg ||\sum_{x_i \in R_j} \nabla_\theta \updownarrow(\cdot)||$. This disparity skews the gradient descent direction toward minimizing majority class errors, thereby shifting the decision boundary toward the minority region and biasing the model toward predicting negative labels. Furthermore, the regularization term $\lambda \nabla_\theta \Omega(\theta)$ may overwhelm the weak gradient signals from R_j, leading to underfitting of rare labels.

From the perspective of statistical learning theory, the generalization performance of a multi-label classifier is jointly influenced by feature dimensionality *d*, total instance size *m*, and the imbalance ratio ρ_j. For a minority label l_j, the generalization error bound can be intuitively expressed as $\mathcal{O}(\sqrt{(\rho_j d)/m})$, which reflects the coupling among these factors. Since $\rho_j = m/|R_j|$, increasing *m* does not necessarily improve minority label learning unless $|R_j|$ grows proportionally. In high-dimensional settings, large *d* coupled with high ρ_j further enlarges the generalization gap, highlighting the intrinsic coupling among *d*, *m*, and ρ_j. This analysis demonstrates the necessity of designing ML-CIL algorithms that explicitly address label class imbalance.

C. Taxonomy of Multi-Label Class Imbalance Learning

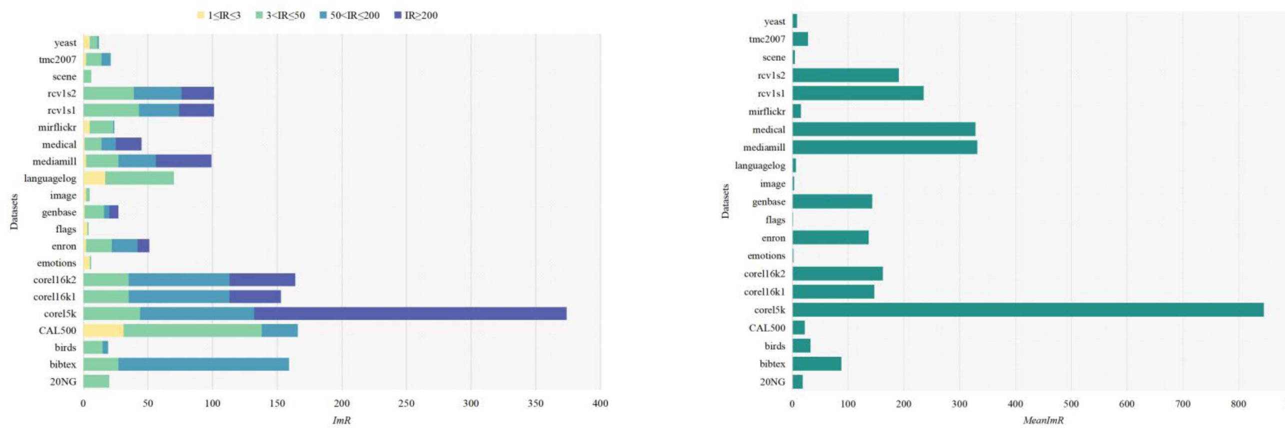
Sampling, cost-sensitive learning, threshold moving, and ensemble learning have been widely used in single-label class imbalance learning [14], [19], [25]. Therefore, researchers have expanded these methods to the ML-CIL scenario through a series of adaptations and improvements. In addition, other adaptive algorithms and new learning techniques have also been used to solve ML-CIL problems and are classified as other single methods. Specifically, sampling methods [21], [22], [26] improve model classification performance by optimizing the label distribution of training data. Cost-sensitive learning [25], [27], [28] is another effective method for handling ML-CIL. It introduces a training cost to make the classifier pay more attention to minority labels. Threshold moving [29], [30], [31], as a post-processing technique, addresses the ML-CIL problem by adjusting decision thresholds or compensating for decision outputs after the classifier has been trained. Other single methods [32], [33], [34] are difficult to classify into any of the above categories. They generally solve ML-CIL problems through classifier adaptation or new learning techniques. Ensemble learning methods [19], [23], [35] integrate two or more of the above methods into a stronger classifier to further improve classification performance.

Solution strategies and label correlation are also two commonly used classification criteria in the ML-CIL literature. Specifically, multi-label algorithms can be broadly categorized into problem transformation and algorithm adaptation approaches based on their underlying solution strategies [4], [14], [36]. Problem transformation approaches aim to adapt data to existing algorithms by converting multi-label datasets to single-label datasets or other representational forms. This enables single-label classifiers to be applied to multi-label learning tasks. Conversely, algorithm adaptation methods modify learning algorithms to directly handle multi-label data by altering the learning process rather than changing the data itself.

Label correlation strategies are typically categorized into first-order, second-order, and higher-order approaches [4], [37]. The first-order strategy treats each label independently, ignoring label correlations [38], [39], [40]. The second-order strategy considers interactions between label pairs [23], [24], [41]. The higher-order strategy captures more complex dependencies among multiple labels [19], [42], [43].

III. Multi-Label Dataset Characterization Metrics

Accurate and comprehensive characterization of multi-label data is essential for advancing research on ML-CIL. It not only reveals and quantifies potential class biases and data distributions, but also provides a theoretical foundation for the design and optimization of ML-CIL algorithms. Compared to single-label data, the characterization of multi-label data is more complex because each instance may be associated with both majority and minority labels, which renders its class assignment ambiguous. Consequently, researchers have developed targeted



(a) *ImR* (b) *MeanImR*

Fig. 2. The imbalance metrics (*ImR* (a) and *MeanImR* (b)) on the 21 common benchmark multi-label datasets.

characterization metrics for ML-CIL from diverse perspectives, as detailed below.

The label imbalance ratio-based metrics [23], [25] are among the most commonly used measures for characterizing multi-label data in ML-CIL. Specifically,

Imbalance ratio (ImR): It quantifies the imbalance of a single label l_j using the ratio of negative to positive instances. A higher value indicates more severe imbalance and a higher risk of positive instances being overlooked.

$$ImR_j = \left|\bar{R}_j\right| / \left|R_j\right| \tag{5}$$

Mean of the ImR (MeanImR): It is the average of *ImR* values and reflects the overall imbalance degree of the dataset.

$$\boldsymbol{MeanImR} = \frac{1}{q}\sum_{j=1}^{q} ImR_j \tag{6}$$

Fig. 2 presents the *ImR* and *MeanImR* values for 21 common benchmark datasets.

If *ImR* and *MeanImR* measure intra-label imbalance, then the following four metrics proposed by Charte et al. [20], [44] are designed to quantify inter-label imbalance:

Imbalance ratio per label (IRLbl): This metric quantifies the degree of imbalance between labels by dividing the number of positive instances of the most frequent label by the number of positive instances of other labels.

$$\boldsymbol{IRLbl_j} = \frac{|R_{\text{Max}}|}{|R_j|} \tag{7}$$

Mean of IRLbl (MeanIR): It is calculated by averaging the *IRLbl* values of all labels and is used to assess the overall degree of imbalance between labels.

$$\boldsymbol{MeanIR} = \frac{1}{q}\sum_{j=1}^{q} IRLbl_j \tag{8}$$

Maximum imbalance ratio (MaxIR): The ratio of the most frequent label to the rarest label. A higher value indicates more extreme label imbalance.

$$\boldsymbol{MaxIR} = \max\nolimits_{l_j \in L} IRLbl_j \tag{9}$$

***Coefficient of variation of IRLbl* (*CVIR*):** It is obtained by calculating the ratio of the standard deviation of the *IRLbl* values of all labels to *MeanIR*. It reflects the variation in inter-label imbalance, with a higher *CVIR* value indicating a greater disparity in the degree of imbalance among labels.

$$\boldsymbol{CVIR} = \frac{1}{MeanIR}\sqrt{\frac{\sum_{j=1}^{q}\left(IRLbl_j - MeanIR\right)^2}{q-1}} \tag{10}$$

In addition, there are some metrics that quantify label co-occurrence (as illustrated in Fig. 3 [45], [46]) and local imbalance from the perspective of label-subset imbalance. Specifically, the ***score of concurrence among imbalanced labels* (*SCUMBLE*)** [47] reflects the co-occurrence among majority and minority labels. A high score indicates significant co-occurrence among labels, making the multi-label learning task more difficult.

$$\boldsymbol{SCUMBLE} = \frac{1}{m}\sum_{i=1}^{m}\left[1 - \frac{1}{\overline{IRLbl_i}}\left(\prod_{l_j \in L} IRLbl_{ij}\right)^{\frac{1}{q}}\right],$$

$$IRLbl_{ij} = \begin{cases} IRLbl_j, & l_j \in y_i \\ 0, & l_j \notin y_i \end{cases} \tag{11}$$

where $\overline{IRLbl_i}$ represents the average imbalance level of the label set corresponding to the instance x_i.

Local imbalance measure (LImb): [21] measures the degree of imbalance in the label subset by comparing the label distribution of an instance with that of its neighboring instances.

$$\boldsymbol{LImb} = \frac{1}{q}\sum_{j=1}^{q}\left(\frac{\sum_{i=1}^{m} C_{ij} I\left(y_{ij} = g_j\right)}{\sum_{i=1}^{m} I\left(y_{ij} = g_j\right)}\right),$$

$$C_{ij} = \frac{1}{k}\sum_{x_u \in N_i^k} I\left(y_{uj} \neq y_{ij}\right) \tag{12}$$

where N_i^k denotes the *k* nearest neighbors of instance x_i, C_{ij} quantifies the local imbalance of instance x_i on label l_j, and $g_j \in \{0,1\}$ indicates which class of label l_j is the minority. Therefore, a larger *LImb* indicates greater learning difficulty.

In addition to the specialized metrics designed to measure ML-CIL, basic characterization methods are also employed. These methods can be broadly divided into two categories:

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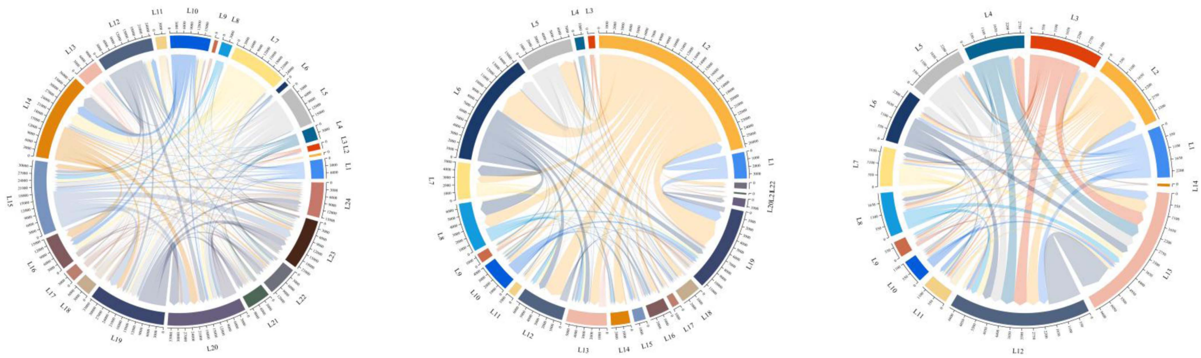

(a) mirflickr (b) tmc2007 (c) yeast

Fig. 3. Concurrence among labels in three common benchmark multi-label datasets.

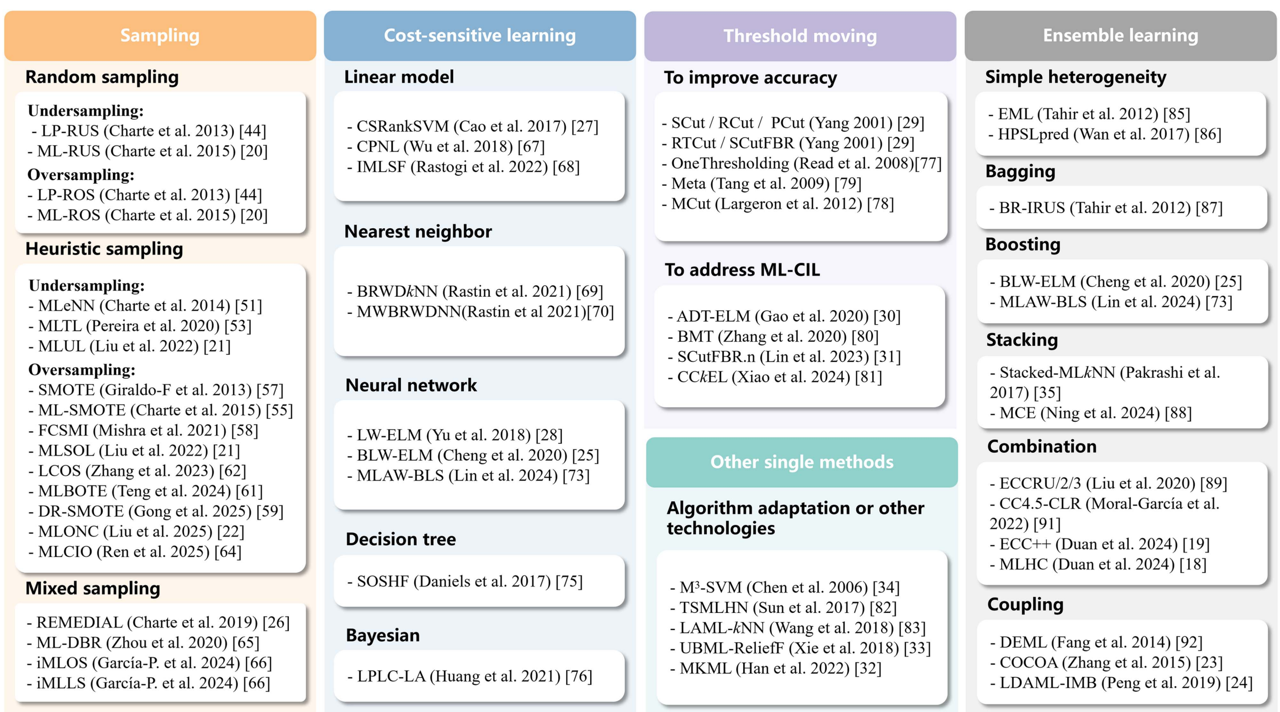

Fig. 4. Classification of methods proposed in the literature to address ML-CIL problems.

quantitative statistics [14], [43], such as the number of instances, features, and labels; and label distribution statistics [23], [43], [48], such as *label cardinality* (*LCard*), *label density* (*LDens*), *distinct label sets* (*DL*), and *the proportion of distinct label sets* (*PDL*).

IV. Approaches for Multi-Label Class Imbalance Learning in Machine Learning

This section provides a systematic review of recent advances in ML-CIL, organized into five main categories based on their solution strategies. Fig. 4 summarizes these approaches. Key information about various algorithms is provided in each subsection (see Tables II–VI). Details of the literature search strategy are provided in the Appendix. Finally, comparative analyses and empirical insights are presented.

A. Sampling

Sampling methods are crucial in class imbalance learning research due to their simplicity, intuitiveness, and efficiency. They are primarily categorized as oversampling and undersampling. Oversampling balances class distributions by creating synthetic minority class instances, while undersampling reduces instances from the majority class. Additionally, sampling methods can also be categorized as random and heuristic, based on how instances are generated or removed. In ML-CIL, sampling strategies remain essential, but the complexity of multi-label datasets limits the applicability of traditional sampling methods, which has motivated the development of specialized sampling approaches for multi-label classification.

In terms of random sampling, Charte et al. [44] proposed the Label Powerset-based Random Oversampling (LP-ROS) and Undersampling (LP-RUS) methods, leveraging the Label

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TABLE II
LIST OF SAMPLING METHODS

No.	Algorithm name	Sampling strategy	Solution strategy	Label correlation	Year
1	SMOTE [57]	Heuristic oversampling	Problem transformation	Not Considered	2013
2	LP-ROS [44]	Random oversampling	Problem transformation	Considered	2013
3	LP-RUS [44]	Random undersampling	Problem transformation	Considered	2013
4	MLeNN [51]	Heuristic undersampling	Algorithm adaptation	Not Considered	2014
5	ML-ROS [20]	Random oversampling	Algorithm adaptation	Not Considered	2015
6	ML-RUS [20]	Random undersampling	Algorithm adaptation	Not Considered	2015
7	ML-SMOTE [55]	Heuristic oversampling	Algorithm adaptation	Not Considered	2015
8	REMEDIAL [26]	Mixed sampling	-	Not Considered	2019
9	MLTL [53]	Heuristic undersampling	Algorithm adaptation	Not Considered	2020
10	ML-DBR [65]	Mixed sampling	-	Not Considered	2020
11	FCSMI [58]	Heuristic oversampling	Problem transformation	Not Considered	2021
12	MLUL [21]	Heuristic undersampling	Algorithm adaptation	Considered	2022
13	MLSOL [21]	Heuristic oversampling	Algorithm adaptation	Considered	2022
14	LCOS [62]	Heuristic oversampling	Problem transformation	Considered	2023
15	MLBOTE [61]	Heuristic oversampling	Algorithm adaptation	Considered	2024
16	iMLOS [66]	Mixed sampling	Algorithm adaptation	Not Considered	2024
17	iMLLS [66]	Mixed sampling	Algorithm adaptation	Not Considered	2024
18	DR-SMOTE [59]	Heuristic oversampling	Algorithm adaptation	Not Considered	2025
19	MLONC [22]	Heuristic oversampling	Algorithm adaptation	Considered	2025
20	MLCIO [64]	Heuristic oversampling	Algorithm adaptation	Considered	2025

TABLE III
LIST OF COST-SENSITIVE LEARNING METHODS

No.	Algorithm name	Specific classifier	Solution strategy	Label correlation	Year
1	CSRankSVM [27]	Linear classification model	Problem transformation	Considered	2017
2	SOSHF [75]	Decision tree	Problem transformation	Considered	2017
3	LW-ELM [28]	Neural network	Algorithm adaptation	Not Considered	2018
4	CPNL [67]	Linear classification model	Problem transformation	Considered	2018
5	BLW-ELM [25]	Neural network	Algorithm adaptation	Not Considered	2020
6	BRWD*k*NN [69]	Nearest neighbor	Problem transformation	Not Considered	2021
7	MWBRWDNN [70]	Nearest neighbor	Problem transformation	Considered	2021
8	LPLC-LA [76]	Bayesian	Algorithm adaptation	Considered	2021
9	IMLSF [68]	Linear classification model	Algorithm adaptation	Considered	2022
10	MLAW-BLS [73]	Neural network	Algorithm adaptation	Considered	2024

TABLE IV
LIST OF THRESHOLD MOVING METHODS

No.	Algorithm name	Design intent	Label correlation	Year
1	SCut / RCut / PCut [29]	Approaches to accuracy	Not Considered	2001
2	RTCut / SCutFBR [29]	Approaches to accuracy	Not Considered	2001
3	OneThresholding [77]	Approaches to accuracy	Not Considered	2008
4	Meta [79]	Approaches to accuracy	Not Considered	2009
5	MCut [78]	Approaches to accuracy	Not Considered	2012
6	ADT-ELM [30]	Approaches to ML-CIL	Not Considered	2020
7	BMT [80]	Approaches to ML-CIL	Not Considered	2020
8	SCutFBR.n [31]	Approaches to ML-CIL	Not Considered	2023
9	CC*k*EL [81]	Approaches to ML-CIL	Considered	2024

TABLE V
LIST OF OTHER SINGLE METHODS

No.	Algorithm name	Solution strategy	Label correlation	Year
1	M^3-SVM [34]	Problem transformation	Not Considered	2006
2	TSMLHN [82]	Algorithm adaptation	Considered	2017
3	LAML-*k*NN [83]	Algorithm adaptation	Not Considered	2018
4	UBML-ReliefF [33]	Problem transformation	Not Considered	2018
5	MKML [32]	Algorithm adaptation	Considered	2022

Powerset (LP) approach [39] to transform multi-label data into single-label data by considering each label combination as a unique class. They identified majority and minority classes based on the number of transformed classes and applied random oversampling or undersampling until reaching a specified percentage. However, LP-based strategies struggle to effectively distinguish between majority and minority classes, and may even fail to sample correctly when label subsets are not significantly imbalanced or the label space is extremely sparse. Therefore, Charte et al. [20] also proposed the Multi-Label-based Random Oversampling (ML-ROS) and Undersampling (ML-RUS) algorithms, which identify majority and minority labels by evaluating the imbalance ratio of each individual label. A label is considered a minority if its *IRLbl* exceeds the *MeanIR*; otherwise, it is classified as a majority label. ML-ROS randomly clones instances that contain minority labels and continuously updates *IRLbl* values to guide further cloning decisions. In contrast, ML-RUS removes instances associated with majority labels but excludes any instances containing minority labels, even if they also contain majority labels.

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TABLE VI
LIST OF ENSEMBLE LEARNING METHODS

No.	Algorithm name	Integrated strategy	Label correlation	Year
1	EML [85]	Simple heterogeneity	Not Considered	2012
2	BR-IRUS [87]	Bagging	Not Considered	2012
3	DEML [92]	Coupling	Considered	2014
4	COCOA [23]	Coupling	Considered	2015
5	Stacked-MLkNN [35]	Stacking	Considered	2017
6	HPSLpred [86]	Simple heterogeneity	Not Considered	2017
7	LDAML-IMB [24]	Coupling	Considered	2019
8	BLW-ELM [25]	Boosting	Not Considered	2020
9	ECCRU/2/3 [89]	Combination	Considered	2020
10	CC4.5-CLR [91]	Combination	Considered	2022
11	MCE [88]	Stacking	Considered	2024
12	MLAW-BLS [73]	Boosting	Considered	2024
13	ECC++ [19]	Combination	Considered	2024
14	MLHC [18]	Combination	Considered	2024

However, these Multi-Label-based (ML-based) strategies still struggle to address the issue of co-occurrence between minority and majority labels. To tackle this, Charte et al. [26] further proposed the Resampling Multi Label Datasets by Decoupling Highly Imbalanced Labels (REMEDIAL) algorithm to alleviate the impact of label concurrence. This approach applies the *SCUMBLE* metric [47] to quantify label co-occurrence for each instance. Instances with high *SCUMBLE* values are split into two new instances: one retaining only minority labels and the other retaining only majority labels. REMEDIAL is versatile and can be integrated with various sampling algorithms [17], [49], [50]. While REMEDIAL effectively addresses the label co-occurrence issue, label decoupling may disrupt original label correlations.

Although random sampling methods are simple to implement and computationally efficient, they overlook potential relationships within the multi-label data, leading to blind instance selection. In extreme scenarios—characterized by high-dimensional features, label sparsity, and complex dependencies—random sampling methods often fail to capture representative instances and may even introduce noise. Consequently, heuristic sampling algorithms, which offer a more strategic and intelligent approach to instance selection or synthesis, have emerged as a significant focus in ML-CIL research in recent years.

Unlike LP-RUS, the Multi Label edited Nearest Neighbor (MLeNN) [51] approach uses eNN [52] and ML-based rules to identify and remove instances that differ from the label sets of neighboring instances and do not contain minority class labels. Similarly, the Multi-Label Tomek Link (MLTL) [53] method employs a strategy inspired by the classic Tomek Link [26] to identify pairs of neighboring instances with significant differences in label sets and remove those that may confuse the class boundaries. The Multi-Label Undersampling based on Local Label Imbalance (MLUL) [21] algorithm is the first to adopt a local perspective by using the "Reverse k-Nearest Neighbors" (RkNN) algorithm [54] to identify harmful instances and comprehensively assess their contributions to the model, ultimately selectively retaining those that are beneficial.

Typically, oversampling methods outperform their undersampling counterparts under similar strategies [20], [55]. This is expected, as undersampling invariably results in the loss of crucial information, irrespective of whether random or heuristic approaches are used. Acknowledging this challenge, researchers have made significant advancements in heuristic-based oversampling techniques in recent years. The Synthetic Minority Oversampling Technique (SMOTE) family is a prominent method among heuristic oversampling techniques [56], and this also holds true for ML-CIL. Giraldo-Forero et al. [57] pioneered the application of SMOTE in the ML-CIL domain and proposed three specific strategies: sampling each label individually based on the binary relevance (BR) [4] principle; sampling only from single-label ("pure") instances; and sampling from all instances associated with minority labels. Experimental results show that only the third strategy is effective, as the others ignore label correlations and use suboptimal seeds. Similarly, the Feature Construction and Smote-based Imbalance handling (FCSMI) [58] method adopts the BR+SMOTE strategy and constructs a new feature space by computing the distances between each instance and minority class instances, thereby enhancing the applicability of SMOTE. In addition, SMOTE based on algorithm adaptation strategies has also achieved significant results, such as the Multilabel Synthetic Minority Over-sampling Technique (ML-SMOTE) [55] algorithm, which employs the ML-based strategy [20] to identify minority labels as seed instances. It then generates the label set for synthetic instances by referencing their neighbors using intersection, union, or majority voting methods. Experiments demonstrate that the third strategy offers superior performance, as it balances the richness and accuracy of label information. The Diversity and Reliability-enhanced SMOTE (DR-SMOTE) [59] algorithm further identifies "higher-quality" seeds among minority instances selected based on ML strategies using adjusted Hamming distance [53]. It then generates features for synthetic instances using a probabilistic approach, with labelsets assigned to match those of the seed instances. This design mitigates overfitting and boundary overflow issues, thereby enhancing the diversity and reliability of synthetic instances. There are other algorithms which also synthesize instances using linear interpolation, even though they do not explicitly employ SMOTE technology. Sadhukhan and Palit [60] propose selecting non-isolated minority class instances as seeds based on the number of RkNN [54] and perform linear interpolation for each label separately. The Multi-Label Synthetic Oversampling based on Local Label Imbalance (MLSOL) [21] approach identifies and selects "difficult-to-learn" seed instances based on local label imbalance, and intelligently determines the label distribution of synthesized instances by considering label differences between seeds and neighboring instances as well as their local types, ensuring that the synthesized instances are both representative and not overfitted. Furthermore, some recent algorithms emphasize the importance of boundary and label correlation, which assist in boundary division, seed selection, and instance synthesis. Specifically, the Multi-Label Borderline Oversampling Technique (MLBOTE) [61] algorithm categorizes instances into interior, self-borderline, and cross-borderline instances based on their distance to the category boundaries, and then applies different oversampling mechanisms to each category using a “divide and conquer” strategy. The Label Correlation Guided Oversampling (LCOS) [62] method defines inner and outer boundaries based on the label correlation matrix learned by the Simple Recurrent Neural Network [63]. These boundaries contain minority class instances that are challenging to classify. The algorithm synthesizes new instances within these regions through linear interpolation based on the distance weights of each instance, ensuring that the synthesized instances retain the same label set as their original instances. The Multi-Label Oversampling with Natural Neighbor and Label Correlation (MLONC) [22] method dynamically determines the number of neighbors for each instance using natural neighbor detection and prioritizes minority class instances near decision boundaries as seeds to generate additional synthetic instances in these regions. The labels of the synthetic instances are then assigned based on label correlation. In addition to linear interpolation, the Ensemble Over-sampling Approach with Chebyshev Inequality and Group Optimization Strategy (MLCIO) [64] utilizes Chebyshev’s inequality to synthesize new instances within an m-standard deviation range around each seed instance and employs an optimized sorting weighting scheme for labelset generation. Compared with traditional linear interpolation techniques, it considers a broader range of influences beyond nearest neighbors. As a result, it models the data distribution more accurately while reducing computational overhead.

In addition, some mixed sampling methods have also demonstrated efficacy for ML-CIL problems. Zhou et al. [65] and García-Pedrajas et al. [66] employed label decoupling and evolutionary computation methods, respectively, to simultaneously undersample the majority class and oversample the minority class.

B. Cost-sensitive Learning

Unlike sampling methods, cost-sensitive learning is a technique that introduces training costs at the classifier level, thereby preserving the original data distribution. Instead of minimizing overall instance error, it focuses on reducing the total misclassification cost. Specifically, this approach imposes higher penalties for misclassifying minority classes to emphasize their importance. In ML-CIL, cost-sensitive learning can adjust misclassification costs at the label level, the instance level, or both.

Cost-sensitive learning algorithms are generally correlated with a specific classification model due to variations in structures and optimization objectives, which require tailored cost terms for each algorithm [19], [28]. In linear classification models, this is often achieved by adjusting the loss function. The Cost Sensitive Ranking Support Vector Machine (CSRankSVM) [27] approach introduces weights based on label combination frequencies into the ranking loss term of RankSVM [41], thereby increasing the penalty for misclassifying minority label combinations. For label combinations with extremely low frequencies, pruning strategies can be employed to further reduce their negative impact on model training. The Cost-sensitive Multi-label Learning Model with Positive and Negative Label Pairwise Correlations (CPNL) [67] incorporates training costs based on SVM penalty terms, while introducing a relaxation coefficient to control classification errors and integrating label correlations into the regularization term to enhance the model’s ability to handle ML-CIL. It is based on the BR [4], and its weight matrix is related to the number of positive and negative instances for each label. Similarly, the Imbalance Multi-label Data Learning with Label Specific Features (IMLSF) [68] method constructs a linear model by integrating squared hinge loss, class imbalance weighting, label correlation regularization, and label-specific feature selection, and is effective in handling the ML-CIL problem.

Cost-sensitivity can also be introduced by adjusting the distance-sensitivity of Nearest Neighbor (NN-based) algorithms. The Binary Relevance Weighted Distance k Nearest Neighbor (BRWDkNN) [69] algorithm enhances the distance metric in kNN through an improved Prototype Weighting (PW) method, making minority class instances more likely to be selected. Based on the BR method, the algorithm then optimizes an objective function—such as maximizing the Macro-F1 score—for each label via gradient descent to determine the appropriate weights. Similarly, the Meta Weighted Binary Relevance Weighted Distance Nearest Neighbor (MWBRWDNN) [70] approach is developed by weighting features through a weighted distance strategy. It is integrated into a meta-learning framework, where a two-layer structure is trained to fully exploit label correlations.

Cost-sensitive improvements for neural networks can either enhance the expected outputs of minority class labels or impose penalties for their misclassification. The Label Weighted Extreme Learning Machine (LW-ELM) method [28], based on Extreme Learning Machines (ELM) [71], improves error tolerance by increasing the expected outputs of minority class labels instead of imposing stronger misclassification penalties. The corresponding weights are determined according to the degree of label imbalance. Although this empirical approach to amplifying minority class outputs is intuitive, it does not always ensure optimality. To overcome this, Cheng et al. [25] proposed the Boosting LW-ELM (BLW-ELM) algorithm, which employs the Boosting framework [72] to iteratively determine appropriate label weights, coupled with a voting mechanism to improve classification results. Similarly, the Multi-label Adaptive Weighted Broad Learning System (MLAW-BLS) [73] incorporates a label weighting matrix into the loss function of the Broad Learning System [74]. It then adaptively and dynamically adjusts label weights during each Boosting iteration based on misclassification feedback, while also considering label imbalance ratios and label correlations. In fact, due to their use of the Boosting framework, both BLW-ELM and MLAW-BLS can also be classified as ensemble learning methods.

In addition, cost-sensitive improvements have also been explored in decision tree (including random forests) models and Bayesian frameworks. The Sparse Oblique Structured Hellinger Forests (SOSHF) [75] performs cost-sensitive clustering based on label frequency, enabling the transformation of multi-label problems into single-label tasks while enhancing the influence of minority labels at split nodes. The Bayesian Model by Exploiting the Local Positive and Negative Label Correlations with Label Awareness (LPLC-LA) [76] improves model performance by jointly considering label-specific features, local label correlations, imbalance weights, and separability weights to adjust the conditional probability of each label within the Bayesian model.

C. Threshold Moving

Threshold moving is a fundamental technique for handling class imbalance and has proven effective in ML-CIL. It is a post-processing strategy that adjusts the decision rule of a trained classifier without altering the data distribution. Generally, existing methods can be categorized into: (1) decision boundary shifting, which adjusts classification thresholds, and (2) decision output compensation, which modifies predictive scores. Specifically, decision boundary shifting typically involves searching for label-specific optimal thresholds on a validation set to replace the default 0.5. By shifting these boundaries toward majority classes, the model expands the decision region for minority labels, thereby alleviating class imbalance bias. In contrast, decision output compensation adjusts the continuous confidence scores based on label prior distributions or label correlations. Instead of altering the discrete decision rule, it aims to reduce the bias toward majority classes by reconstructing a more balanced posterior distribution.

Early studies mainly focused on improving predictive performance rather than explicitly addressing ML-CIL. Representative methods include score-based local optimization (Scut), ranking-based thresholding (RCut), proportion-based allocation (PCut), RCut with Precision–Recall Trade-off (RTCut), SCut with Fallback Rank (SCutFBR) [29], OneThresholding [77], Maximum Cut Thresholding (MCut) [78], and the meta-learning-based thresholding method Meta [79]. All these methods fall into the category of decision boundary shifting. Although they were not specifically designed for ML-CIL, they provide a valuable theoretical foundation.

Since 2020, some threshold moving methods have been developed for ML-CIL scenarios. For instance, the Adaptive Decision Threshold-based Extreme Learning Machine (ADT-ELM) [30], Bagging with Moved Thresholds (BMT) [80], and SCut with Fallback Rank using Two-layer Cross-validation (SCutFBR.n) [31] remain within this paradigm. Specifically, ADT-ELM optimizes label-wise thresholds using Macro-F and Micro-F measures; BMT aligns predicted and training label distributions via threshold adjustment; and SCutFBR.n refines threshold selection through cross-validation and smoothing. These are also decision boundary shifting methods. In recent years, decision output compensation methods have gradually emerged. A typical example is the Compensation-based Correlated *k*-labelset method (CC*k*EL) [81], which partitions labels into *k* highly correlated subsets based on Jaccard similarity. After encoding and decoding operations, it addresses the multi-label class imbalance problem through a fast decision output compensation strategy.

Current threshold-moving methods for ML-CIL still exhibit limited diversity, among which CC*k*EL is one of the few approaches that explicitly consider label correlations.

D. Other Single Methods

In addition, there are non-ensemble methods that cannot be directly categorized into the aforementioned paradigms, which are collectively referred to as other single methods.

Classifier adaptation methods include the Two-stage Multi-Label Hypernetwork (TSMLHN) proposed by Sun and Lee [82], where the first stage employs a multi-label hypernetwork (MLHN) for preliminary label predictions. In the second stage, label correlations between majority and minority labels are explicitly modeled to further optimize the learning of imbalanced labels. The Locally Adaptive Multi-Label *k*-Nearest Neighbor (LAML-*k*NN) [83] algorithm enhances ML-*k*NN by clustering the training data and adjusting the posterior probabilities of label assignments based on cluster membership information of test instances. The Multi-Kernel Multi-Label (MKML) [32] method introduces three kernel modules to model the global structure, label correlations, and class imbalance, which are jointly optimized within a unified multi-kernel learning framework to improve model performance in ML-CIL contexts. In addition to classifier adaptation methods, the Improved Multi-label Relief Feature Selection Algorithm for Unbalanced Datasets (UBML-ReliefF) [33] enhances the feature selection algorithm by incorporating contribution values into the feature weight update formula, thereby mitigating the impact of uneven sample label selection. The Min-Max Modular Network SVM (M^3-SVM) [34] addresses the ML-CIL issue by breaking it down into smaller, more balanced binary classification subproblems through various sub-task decomposition strategies. Following the min-max modular network principle [84], the outputs of these sub-classifiers are combined to produce the final predictions.

E. Ensemble Learning

Ensemble learning methods, which integrate multiple base learners or ML-CIL strategies, demonstrate significant performance benefits in ML-CIL. By leveraging the accuracy and diversity of individual learners, these methods provide a more comprehensive consideration of skewed label distributions and potential label dependencies, thereby enhancing the recognition of minority class labels. The diversity inherent in ensemble learning also improves model robustness, mitigates overfitting in imbalanced multi-label data, and ensures more stable and accurate predictions. Ensemble learning has been widely employed to address ML-CIL challenges [23].

The most straightforward approach is heterogeneous ensemble learning. The Ensemble of Multi-label Classifiers (EML) [85] combines five distinct multi-label classifiers using average fusion, weighted combination, and adaptive thresholds. Similarly, the Ensemble Multi-Label Classifier for Human Protein Subcellular Location Prediction (HPSLpred) [86] integrates twelve binary classifiers, employing sampling and threshold moving to tackle the ML-CIL issue in predicting human protein subcellular locations. Bagging-based methods, such as the Binary Relevance Inverse Random Under-sampling (BR-IRUS) method [87], perform inverse random undersampling for each label to generate multiple subsets, each containing fewer majority instances than minority ones. These subsets are trained separately, and their predictions are combined through majority voting. Boosting-based methods, such as BLW-ELM [25] and MLAW-BLS [73], which were mentioned earlier, are also prevalent. These approaches leverage classifier diversity and label correlations to improve minority class label learning.

Beyond conventional ensembles, meta-learning frameworks have recently been applied to ML-CIL by learning from base classifier outputs to capture label correlations and imbalance patterns. Methods such as Stacked-ML*k*NN [35] and Meta-confidence Ensemble (MCE) [88] employ meta-level mechanisms to refine predictions and calibrate outputs for minority labels. Functionally, meta-learning in ML-CIL primarily enables dynamic weighting and calibration rather than explicit base classifier selection.

Recent research in ML-CIL has focused on ensemble learning methods that combine classic multi-label strategies with single-label imbalanced classifiers. These methods are easy

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE VII
ADVANTAGES AND DISADVANTAGES OF DIFFERENT CATEGORIES OF ML-CIL METHODS

Approach	Advantages	Disadvantages
Sampling	- Not restricted by classifiers - Relatively simple and easy to implement	- Need to modify the original data distribution - May delete important information or introduce noise
Cost-sensitive learning	- High computing efficiency - Original data distribution remains unchanged	- Limited by specific classifiers - Difficulty in allocating appropriate weights
Threshold moving	- High computing efficiency - Relatively simple and easy to implement	- Limited by soft output classifier - Prone to overfitting
Other single methods	- New design concepts - Effective in a certain context	- Requires extensive knowledge of specific classifiers and problem domains
Ensemble learning	- Superior performance and high robustness - Integrating the advantages of multiple methods	- High model complexity - High computational overhead

to implement, effectively combining the advantages of both approaches, and yield remarkable results. The Ensemble of Classifier Chains with Random Undersampling (ECCRU) [89] integrates ECC [90] with RUS to address ML-CIL by modeling chain label correlations and balancing class distributions in binary training sets. Enhanced versions, ECCRU2 and ECCRU3, have also been proposed to improve instance utilization. The ECC++ algorithm family [19] further extends the ECC by incorporating three classic class imbalance strategies: sampling, cost-sensitive learning, and threshold moving. The Multi-label Learning based on Hierarchical Clustering (MLHC) [18] employs hierarchical clustering of the label space via Jaccard similarity, splitting labelsets into subsets of up to three labels. It transforms the multi-label learning into a multi-class learning using binary encoding, applies a multi-class imbalanced learning algorithm for training and prediction, and decodes the results into a predicted multi-label distribution. Moral-García et al. [91] proposed an improved version of C4.5 (Credal C4.5) based on imprecise probability theory and explored its application within the Calibrated Label Ranking (CLR) framework.

Finally, classifier coupling-based ensemble methods have been explored, such as the Cross-coupling Aggregation (COCOA) [23]. It constructs a binary imbalanced learner for each label and randomly selects other labels to combine with the current label, training a multi-class classification imbalanced learner. The predictions from these binary and multi-class learners are then coupled to generate the final label prediction. However, the random label coupling strategy in COCOA has limitations, which are addressed by the Latent Dirichlet Allocation for Multi-label learning with Imbalance Correction (LDAML-IMB) [24] framework. Specifically, it treats the labelset of an instance and a label as an article and a word, respectively, using the LDA model to reflect label correlation. This allows only highly correlated labels to be coupled, thereby improving model performance. Similarly, the Dealing with Labels Imbalance by Entropy for Multi-Label Classification (DEML) [92] method constructs classifiers for each label using the BR method, and then applies entropy as a metric for random sampling to create relatively balanced label subsets. These subsets are transformed using LP and used to train classifiers. The final result is obtained by integrating the predictions from BR and LP.

F. Comparative Analysis and Practical Guidelines

Although numerous methods have been developed for ML-CIL, their practical applicability remains unclear due to the lack of a unified comparative perspective. This section presents a concise cross-method analysis and practical guidelines for method selection, as summarized in Table VII.

Specifically, sampling methods improve classification performance by optimizing data distributions. They are easy to implement and can be integrated with various multi-label algorithms. However, they may alter the original data distribution, and challenges arise in synthetic label assignment and sampling boundary determination. Cost-sensitive learning introduces training costs to emphasize minority labels without changing the data distribution, but it requires algorithmic improvements. How to incorporate costs and assign appropriate weights remains challenging. Threshold moving is a post-processing technique. It is limited to classifiers with soft outputs and is prone to overfitting. Other single methods rely on classifier adaptation or new technique transfer, which makes them difficult to unify under a common paradigm. Ensemble learning combines multiple approaches to further improve performance, but it often incurs high computational complexity.

To further support the above analysis, we conduct illustrative experiments on representative methods from different types. As shown in Table VIII, the effectiveness of different methods varies with the degree of class imbalance. Sampling methods are more suitable for mild to moderate imbalance because they generate high-quality minority instances to mitigate the issue. However, they perform poorly under extreme imbalance, as they can introduce significant noise and distort the original data distribution. In contrast, cost-sensitive learning is generally more stable in highly imbalanced scenarios because it modifies the loss function weights while preserving the original data distribution, though its success depends heavily on the cost design. Threshold moving offers a lightweight solution without retraining. However, its performance is often limited, especially under severe imbalance, due to unreliable probability estimations or overfitting. Other single methods rely on strong assumptions about classifiers or problem structures and are difficult to unify within a general paradigm; therefore, they are not included in this comparison. Ensemble methods are robust across various scenarios but come with significantly higher computational costs.

From the perspective of dataset scale, the computational complexity of ML-CIL methods is mainly affected by the number of instances, labels, and features. Sampling methods often consider the relative relationships among neighboring samples, making them sensitive to the number of instances and features. Cost-sensitive learning usually introduces instance-level or label-level weights during training. Therefore, its complexity mainly increases with the number of instances and labels, while also depending on the scale sensitivity of the underlying classifier. The complexity of threshold moving methods mainly depends on the threshold calibration strategy. For ensemble learning methods, the complexity depends not only on the scale sensitivity and number of base learners, but also on the specific ensemble

TABLE VIII
PERFORMANCE COMPARISON OF REPRESENTATIVE ML-CIL METHODS (MACRO-F)

Datasets (m^*, d, q, $MeanIR$)	Sampling			Cost-sensitive learning		Threshold moving		Ensemble learning	
	ML-SMOTE [55]	MLSOL [21]	MLONC [22]	LW-ELM [28]	BRWD*k*NN [69]	ADT-ELM [30]	CC*k*EL [81]	COCOA [23]	ECC++ [19]
flags (193, 19, 7, 1.86)	0.4976	0.4917	0.5473	0.4717	0.4638	0.5116	0.4944	0.5250	**0.5525**
emotions (593, 72, 6, 2.32)	0.5111	0.5310	**0.5501**	0.4166	0.4213	0.5016	0.4375	0.4922	0.5290
Image (2000, 294, 5, 3.16)	0.5969	0.5753	0.5825	0.6031	0.5744	0.4873	0.4217	0.4554	**0.6183**
scene (2407, 294, 6, 4.66)	0.6826	0.6818	0.6578	0.7025	0.6939	0.6081	0.5816	0.5737	**0.7612**
CHD_49 (555, 49, 6, 6.58)	0.4445	0.4484	0.4662	0.4767	**0.4911**	0.4271	0.4153	0.4858	0.4624
yeast (2417, 103, 14, 8.58)	0.3797	0.3793	0.3820	0.4087	**0.4416**	0.4197	0.4029	0.3972	0.4393
VirusPseAAC (207, 294, 6, 8.62)	0.3867	0.3640	0.3705	0.3695	0.3117	0.1865	0.1367	**0.3954**	0.3151
PlantPseAAC (978, 440, 12, 21.88)	0.1165	0.1242	0.1189	0.1189	0.1349	0.0910	0.0736	0.1474	**0.1511**
birds (645, 260, 19, 32.86)	0.0889	0.1010	0.0641	0.0641	0.0754	0.0578	0.0523	**0.1125**	0.0722
genbase (662, 1185, 27, 143.46)	0.5919	0.5967	0.5682	0.6075	0.6176	0.4973	0.4251	0.6213	**0.7031**
Average Running Times (s)	0.7925	0.2778	0.1605	0.0430	1.3081	0.0681	0.1445	2.3102	4.4391

Here, m^* denotes the number of instances in each dataset, i.e., $m^* = m + p$

strategy. For example, combination-based and coupling-based methods are usually more sensitive to the number of labels because they need to train one or more classifiers for each label or label group. In addition, methods that consider label correlations may incur additional complexity as the number of labels increases.

The choice of ML-CIL methods should consider both data characteristics and practical constraints. Sampling methods are preferred for mild to moderate imbalance to enhance minority class representation, while cost-sensitive methods provide a more stable alternative for severe imbalance. Threshold moving is suitable for scenarios requiring low computation or post-hoc adjustments. Other single methods can be considered if prior knowledge of the classifier is available, while ensemble methods are recommended when performance is the priority and computational resources are sufficient. Actually, no single strategy is universally optimal, and combining multiple techniques may provide a better balance. For example, sampling and cost-sensitive learning can be combined in a complementary manner. Sampling increases the exposure of minority labels during training, whereas cost-sensitive learning adjusts their contribution to model optimization. This combination can reduce over-reliance on synthetic samples, lower the risk of noise introduction, and alleviate optimization instability caused by excessive weighting. In addition, ensemble methods based on computationally efficient base models can achieve a better balance between model robustness and computational cost.

V. APPROACHES FOR MULTI-LABEL CLASS IMBALANCE LEARNING IN DEEP LEARNING

In recent years, with the rapid advancement of deep learning [93], the ML-CIL problem has significantly affected the training of deep neural networks (DNNs) [94] and has increasingly attracted attention from researchers [95]. Unlike traditional methods, DNNs can effectively extract high-level semantic features from complex and unstructured data [96], [97], [98], which further increases the challenges of ML-CIL in deep learning datasets. Fortunately, in addition to the aforementioned strategies—which have already been shown to be effective in deep learning—more advanced methods that leverage the feature learning capacity of neural networks have demonstrated even more promising results. Current deep learning methods for ML-CIL can be broadly categorized into three levels: (1) data-level strategies that manipulate training data distributions via sampling; (2) loss-level modifications that recalibrate gradient contributions without altering the network backbone, e.g., cost-sensitive learning; and (3) network-level strategies that adapt feature representations or redesign network architectures. While early methods often relied on a single strategy, tackling increasingly complex datasets now demands a synergistic integration of these three levels to achieve more promising and robust results. Fig. 5 summarizes the aforementioned deep ML-CIL methods.

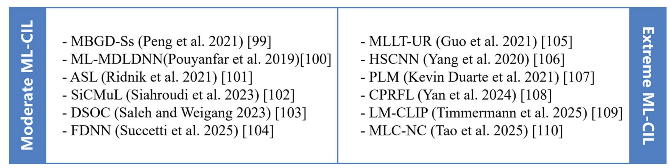

Fig. 5. Classification of ML-CIL methods in the context of deep learning.

A. Moderate Class Imbalance

Early deep ML-CIL approaches often rely on single-level strategies and are primarily designed for data with moderate class imbalance. Specifically, data-level strategies serve as the starting point. These methods typically balance the distribution of the training data through sampling strategies. For instance, the Mini-Batch Gradient Descent with Stratified Sampling (MBGD-Ss) [99] employs the MBGD strategy for dynamic hierarchical sampling, where label combinations and individual labels are used as hierarchical sampling units.

Loss-level methods focus on modifying the optimization objective by adjusting the loss function or training weights. As a form of cost-sensitive learning, these methods do not require modifications to the network architecture. For instance, the Multi-label Classification with Imbalanced Classes by Fuzzy Deep Neural Networks (ML-MDLDNN) [100] adaptively assigns weights to each class in DNNs to address the ML-CIL problem under multimodal scenario. Ridnik et al. [101] proposed the Asymmetric Loss for Multi-Label Classification (ASL) method, which employs asymmetric focusing to adjust the learning difficulty between positive and negative instances, thereby amplifying the gradient contribution of positive instances. Concurrently, an asymmetric probability bias is used to discard easily classified negative instances and address potential labeling errors, reducing training bias toward the majority class.

Although these methods have been shown to be effective in deep ML-CIL, they do not fully exploit the structural capacity of DNNs. Network-level strategies address this limitation by leveraging architectural design or feature learning to capture label-specific information. For example, the effective single-label classifier to solve multi-label data classification (SiCMuL) [102] addresses catastrophic forgetting by employing orthogonal matrices to create label-specific contexts and sharing network weights to learn label correlations. It utilizes neighborhood graph undersampling to eliminate redundant or uninformative samples, effectively mitigating noise while tackling the deep ML-CIL problem. The Deep Self-Organizing Cube (DSOC) [103] enhances the model's learning and recognition capabilities for minority class labels by constructing a self-organizing pooling layer and a multi-dimensional hypercube classifier, combined with feature selection and probability compensation modeling. Succetti et al. [104] introduce a solution utilizing fuzzy deep neural networks (FDNN) to address class imbalance bias in deep ML-CIL. By integrating fuzzy logic to manage label overlap and uncertainty, FDNN enhances the learning of minority class features. It incorporates the Gated Recurrent Units (GRU) layers for capturing time-dependent relationships and leverages the Symbolic Aggregation Approximation (SAX) to expedite training.

B. Extreme Class Imbalance

As the disparity in label frequencies widens, the complexity of deep ML-CIL problems increases significantly. Many deep multi-label datasets exhibit an extreme long-tailed distribution: a few head labels dominate with numerous instances, while most tail labels are extremely rare. This imbalance amplifies the model's dependency on head labels, severely impairing its ability to recognize tail labels. Traditional ML-CIL methods often underperform or fail in these scenarios. Consequently, an increasing number of deep learning approaches have been developed to address multi-label long-tail (ML-LT) problems.

Current ML-LT methods are mainly optimized through the coordinated integration of data-level, loss-level, and network-level strategies based on the characteristics of ML-LT data, rather than relying on a single strategy. Specifically, the Long-Tailed Multi-Label Visual Recognition by Collaborative Training on Uniform and Re-balanced Samplings (MLLT-UR) [105] constructs a dual-branch network that is trained separately on uniformly sampled and resampled data and introduces a novel cross-branch loss function to maintain prediction consistency between the two branches. This approach improves performance on tail labels while preserving accuracy on head labels. Yang et al. [106] introduce a Hybrid-Convolutional Neural Network (HSCNN). This algorithm employs a general network for head labels and utilizes small-sample techniques for tail labels. It integrates a multi-task architecture, category-specific similarity metrics, and targeted sampling to jointly optimize both head and tail labels. The Partial Label Masking for Imbalanced Multi-label Classification (PLM) [107] method employs partial label masking by randomly masking labels during loss calculation to minimize the KL divergence between the predicted and true label distributions. It further adjusts category proportions dynamically based on network performance, thereby improving recall for minority classes and precision for frequent classes.

Additionally, the necessity for multi-strategy fusion has also driven the field toward advanced feature learning and pre-training paradigms. For instance, the Category-Prompt Refined Feature Learning (CPRFL) [108] leverages pretraining of the CLIP text encoder to establish semantic associations between head and tail class labels. It introduces a progressive dual-path backpropagation strategy to mitigate visual-semantic domain bias. The model also employs asymmetric loss to suppress negative instances across all classes, thereby enhancing overall recognition performance. Similarly, the Long-tailed Multi-label Contrastive Language Image Pretraining (LM-CLIP) [109] method integrates CLIP's contrastive loss with an enhanced balanced asymmetric loss to optimize performance across both head and tail labels in long-tail datasets, employing dynamic sampling and class-weighting strategies. The ML-LT problem can also be addressed at the feature level, a growing trend regarded as a more advanced approach. Long-Tailed Multi-Label Image Classification Through the Lens of Neural Collapse (MLC-NC) [110] introduces the Neural Collapse theory. By utilizing Fixed Equiangular Tight Frame (ETF) label embeddings and collapse calibration techniques, this method refines category feature learning, enhancing the discriminative power of all categories while effectively addressing co-occurrence and interdependence issues between head and tail labels.

C. Summary and Discussion

Existing deep learning approaches for ML-CIL can be broadly categorized into data-level, loss-level, and network-level strategies, and each route presents specific advantages and inherent limitations. Data-level strategies directly balance the label distribution and are simple to implement in practical applications. However, they easily cause overfitting on minority classes and often discard valuable information or introduce noise. Loss-level strategies adjust the gradient contributions of different instances or labels during the training process. These methods do not increase the structural complexity of the network and are computationally efficient. However, their performance is sensitive to hyper-parameter settings, and they are often limited in capturing complex label correlations. At the network-level, strategies generally focus on either modifying network architectures or adapting feature learning processes. Methods that modify architectures often design specific network modules to decouple the learning of head and tail labels or capture label dependencies. Meanwhile, feature learning methods map raw inputs into highly discriminative spaces to construct robust representations for tail labels. Both directions significantly improve the recognition accuracy of minority labels. Nevertheless, these network-level approaches typically introduce high computational costs and require complex designs.

In practice, method selection depends on computational cost, label complexity, and the severity of imbalance. For moderate imbalance, data-level or loss-level methods (e.g., ASL) provide effective solutions without changing the base network or adding much computational burden. However, for datasets with complex label correlations, network-level strategies (e.g., SiCMuL, DSOC) are more suitable, as they better capture feature relationships despite higher computational cost. Furthermore, for extreme ML-LT datasets, a single strategy is generally insufficient. Researchers should use hybrid methods (e.g., MLLT-UR, HSCNN) that combine sampling, specific loss functions, and advanced network architecture design. Finally, when tail labels have extremely few samples, pre-trained models (e.g., LM-CLIP) provide a promising direction by leveraging external semantic knowledge to enhance feature learning.

Overall, deep learning approaches for ML-CIL have evolved from single-strategy methods to multi-strategy integration. While moderate ML-CIL problems can often be effectively

TABLE IX
LIST OF PERFORMANCE EVALUATION METRICS

Categories	Metrics and Definitions	Interpretations	Remarks	※
Instance-based measures	$\boldsymbol{Precision} = \frac{1}{p}\sum_{i=1}^{p}\frac{\lvert y_i \cap \hat{y}_i\rvert}{\lvert \hat{y}_i\rvert}$	Measures the average proportion of correctly predicted labels among all predicted labels for each instance.	It evaluates the reliability of positive label predictions and reflects the model's ability to reduce false positives.	↑
	$\boldsymbol{Recall} = \frac{1}{p}\sum_{i=1}^{p}\frac{\lvert y_i \cap \hat{y}_i\rvert}{\lvert y_i\rvert}$	Measures the average proportion of correctly predicted labels among all actual labels for each instance.	It evaluates the completeness of positive label identification and reflects the model's ability to reduce false negatives.	↑
	$\boldsymbol{F-measure} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$	Measures the harmonic mean of instance-based *Precision* and *Recall*.	It balances prediction reliability and label identification completeness.	↑
	$\boldsymbol{Accuracy} = \frac{1}{p}\sum_{i=1}^{p}\frac{\lvert y_i \cap \hat{y}_i\rvert}{\lvert y_i \cup \hat{y}_i\rvert}$	Measures the average overlap between the predicted label set and the actual label set for each instance.	It reflects the overall consistency between predicted and actual label sets.	↑
	$\boldsymbol{Subset\ Accuracy} = \frac{1}{p}\sum_{i=1}^{p} I(y_i = \hat{y}_i)$	Measures the proportion of instances whose predicted label set exactly matches the actual label set.	It is a strict metric because any instance prediction that fails to match the actual label set does not contribute to the score.	↑
	$\boldsymbol{Hamming\ Loss} = \frac{1}{p}\sum_{i=1}^{p}\frac{1}{q}\lvert y_i \,\Delta\, \hat{y}_i\rvert$	Measures the average proportion of misclassified labels for each instance.	It considers both false negatives and false positives, but may underestimate prediction errors when the label space is sparse.	↓
Label-based measures	$\boldsymbol{Macro-F} = \frac{1}{q}\sum_{j=1}^{q} 2 \times \frac{Precision_j \times Recall_j}{Precision_j + Recall_j}$	Calculates the F-measure for each label separately and then takes the average over all labels.	It assigns equal weight to each label, making it more sensitive to minority labels.	↑
	$\boldsymbol{Micro-F} = 2 \times \frac{\overline{Precision} \times \overline{Recall}}{\overline{Precision} + \overline{Recall}}$	Calculates the F-measure after averaging the confusion-matrix elements, including *TP*, *FP*, and *FN*, across all labels.	It reflects overall classification performance, but majority labels may contribute more to the final score.	↑
Ranking-based measures	$\boldsymbol{OneError} = \frac{1}{p}\sum_{i=1}^{p} I(l_{\arg\max_j \hat{y}_{ij}^{*}} \notin y_i)$	Measures the proportion of instances whose top-ranked predicted label is not included in the true label set.	It reflects the correctness of the model's top-ranked prediction, but it ignores the prediction quality of the remaining labels.	↓
	$\boldsymbol{Coverage} = \frac{1}{p}\sum_{i=1}^{p}(\max_{l_j \in y_i}\hat{y}_{ij}^{\mathrm{Rank}} - 1)$	Measures how many top-ranked labels are needed, on average, to cover all relevant labels for each instance.	It reflects whether all relevant labels are ranked sufficiently high, but ignores their internal ranking order.	↓
	$\boldsymbol{Ranking\ Loss} = \frac{1}{p}\sum_{i=1}^{p}\frac{1}{\lvert y_i\rvert\lvert \bar{y}_i\rvert}\sum_{l_j \in y_i}\sum_{l_k \in \bar{y}_i} I(\hat{y}_{ij}^{\mathrm{Rank}} \geq \hat{y}_{ik}^{\mathrm{Rank}})$	Measures the average proportion of irrelevant labels ranked above relevant labels for each instance.	It reflects whether relevant labels are placed higher in the ranking than irrelevant labels.	↓
	$\boldsymbol{Average\ Precision} = \frac{1}{p}\sum_{i=1}^{p}\frac{1}{\lvert y_i\rvert}\sum_{l_j \in y_i}\frac{\sum_{l_k \in y_i} I(\hat{y}_{ik}^{\mathrm{Rank}} \leq \hat{y}_{ij}^{\mathrm{Rank}})}{\hat{y}_{ij}^{\mathrm{Rank}}}$	Measures the average proportion of labels ranked above a particular label l_j that are actually in y_i.	It reflects whether relevant labels appear early and are densely concentrated near the top of the ranking.	↑

Here, ↑(↓) in the ※ column indicates that a larger (smaller) value is better, and Δ denotes the symmetric difference between two sets.

addressed with a single strategy, extreme long-tailed scenarios require a combined approach.

VI. Performance Evaluation for Multi-Label Class Imbalance Learning Methods

Performance evaluation of ML-CIL methods is more complex than that of single-label class imbalance learning. Since each instance is associated with multiple labels at the same time, traditional single-label evaluation metrics cannot be directly applied to ML-CIL. Therefore, specially designed or adapted evaluation metrics are needed to accurately assess the performance of ML-CIL methods.

A. Performance Evaluation Metrics

This section systematically reviews existing evaluation metrics for ML-CIL and analyzes their definitions, applicability, and limitations. In general, these metrics can be categorized into three types: instance-based metrics [111], label-based metrics [112], and ranking-based metrics [113].

Specifically, instance-based metrics first measure the difference or consistency between the predicted label set and the actual label set for each instance. The final metric value is then obtained by averaging the results over all instances. Therefore, these metrics reflect the quality of label prediction from the instance perspective. They usually include *Precision*, *Recall*, *F-measure*, *Accuracy*, *Subset Accuracy*, and *Hamming Loss*. Label-based metrics can also be used to calculate *Precision*, *Recall*, *F-measure*, and other related metrics. Unlike instance-based metrics, they treat each label as a separate binary classification problem and obtain the final metric value through macro or micro averaging. Macro averaging computes the metric for each label and then averages the results, so it is more sensitive to minority labels. In contrast, micro averaging aggregates the prediction results of all labels before computing the metric, thus reflecting overall prediction performance. To distinguish them from instance-based metrics, label-based metrics usually indicate the averaging strategy in their names, such as macro-averaged F-measure (*Macro-F*) and micro-averaged F-measure (*Micro-F*). Ranking-based metrics evaluate the order of labels according to the confidence scores produced by ML-CIL classifiers, rather than relying on binary prediction results. This is valuable for ML-CIL because some relevant labels, especially minority labels, may still obtain high ranking positions even if they are not selected by the decision threshold. Common ranking-based metrics, such as *One Error*, *Coverage*, *Ranking Loss*, and *Average Precision*, are used to evaluate the ranking quality of ML-CIL methods from different perspectives.

Table IX summarizes representative evaluation metrics and presents their definitions, interpretations, and remarks. The definitions of $Precision_j$, $Recall_j$, $\overline{Precision}$ and $\overline{Recall}$ are provided as follows, where TP_j, FP_j, FN_j, and TN_j denote the numbers of true positives, false positives, false negatives, and true negatives for label l_j, respectively.

$$Precision_j = \frac{TP_j}{TP_j + FP_j} \tag{13}$$

$$Recall_j = \frac{TP_j}{TP_j + FN_j} \tag{14}$$

$$\overline{Precision} = \frac{\frac{1}{q}\sum_{j=1}^{q} TP_j}{\frac{1}{q}\sum_{j=1}^{q} TP_j + \frac{1}{q}\sum_{j=1}^{q} FP_j} \tag{15}$$

$$\overline{Recall} = \frac{\frac{1}{q}\sum_{j=1}^{q} TP_j}{\frac{1}{q}\sum_{j=1}^{q} TP_j + \frac{1}{q}\sum_{j=1}^{q} FN_j} \tag{16}$$

B. Sensitivity Analysis and Selection Guidelines

A critical challenge in ML-CIL is that different evaluation metrics exhibit varying degrees of sensitivity to class imbalance. The choice of evaluation metric can significantly affect the interpretation of ML-CIL performance.

Specifically, instance-based metrics assign equal weight to each instance and therefore reflect the overall prediction quality of the label set of each instance. However, they differ in their sensitivity to class imbalance. *Precision* is more sensitive to false positive labels, whereas *Recall* focuses on minimizing false negatives, which is important for identifying minority labels. *F-measure* provides a balanced evaluation of *Precision* and *Recall*. *Subset Accuracy* requires a completely correct label set, but it is often too strict in ML-CIL. *Hamming Loss* evaluates each instance-label pair independently and is therefore relatively stable. However, it may be dominated by numerous negative label assignments and thus underestimate the difficulty of identifying minority labels. Label-based metrics are strongly influenced by the averaging strategy. Macro-averaged metrics assign equal importance to each label, and are therefore more suitable for revealing whether minority labels are effectively learned. By contrast, micro-averaged metrics are computed from all label decisions and therefore mainly reflect the overall classification performance across labels. Accordingly, macro-averaged and micro-averaged results usually need to be considered together when evaluating an ML-CIL method. Furthermore, ranking-based metrics evaluate the relative ordering of prediction probabilities rather than relying on hard decision thresholds. These metrics can accurately measure whether a model gives a higher rank to a true minority label than to an irrelevant one. Therefore, they provide an important evaluation perspective when the predicted probabilities are highly skewed.

In practice, researchers should select evaluation metrics according to the specific application scenario. Instance-based metrics are usually more appropriate when each instance is equally important in evaluation or the evaluation focuses on the prediction quality of the label set of each instance. For label-based metrics, micro-averaging is usually preferred when greater attention is paid to the overall classification performance over all label decisions. In contrast, in scenarios where prediction errors on minority labels may lead to higher costs, such as medical diagnosis or rare fault detection, macro-averaging is usually preferred, because it ensures that the performance on minority labels receives sufficient attention. In addition, ranking-based metrics are particularly suitable for scenarios that emphasize label ranking, such as information retrieval or recommendation systems. In summary, no single metric can fully reflect the performance of ML-CIL methods. For a fair and comprehensive comparison, it is generally recommended to report results from multiple perspectives. Such an evaluation protocol can reduce the risk of biased conclusions and provide more convincing evidence for the effectiveness of ML-CIL methods.

VII. Datasets and Software tools

Most ML-CIL studies rely on publicly available benchmark datasets from repositories such as UCI [114], Kaggle [115], and OpenML [116]. Despite their timely updates, these datasets often exhibit inconsistent formats and varying quality. Consequently, they require extensive preprocessing, which limits their direct compatibility with ML-CIL algorithms. To address these issues, a number of standard multi-label learning software tools have been developed to provide integrated datasets, including MULAN [117] and MEKA [118] for Java, mldr package [119] for R, scikit-multilearn [120] for Python, and MLC Toolbox [121] for MATLAB. These integrated tools are highly convenient and suitable for direct use by most researchers. However, their reliability depends heavily on continuous maintenance by their original developers. In addition, they often rely on diverse programming environments and configurations, which complicates the integration of new datasets and algorithms and hinders fair cross-platform comparisons. To further improve accessibility, some research projects have consolidated these datasets. KDIS Lab extends the datasets from Mulan and Meka, while the Cometa [122] platform integrates almost all the multi-label datasets from the aforementioned software tools. However, most of these repositories primarily focus on data collection and provide limited support for learning algorithms. Meanwhile, some repositories such as LABIC, CLUS [123], and XML [124] offer domain-specific datasets and tools. Although these resources are highly effective in their target domains, researchers rarely use them to evaluate general ML-CIL algorithms. Charte [125] provided a systematic review of the aforementioned multi-label datasets and software tools, indicating that most platforms only incorporate multi-label algorithms but lack support for ML-CIL algorithms. Although purpose-built tools for ML-CIL have begun to emerge—such as the R-based multi-label sampling method proposed by Rivera et al. [126]—their number and coverage remain limited.

With the advancement of deep learning, research on ML-CIL has increasingly shifted toward large-scale, high-dimensional benchmarks with richer semantic complexity. The selection of these datasets is typically determined by the specific objectives, data characteristics, and imbalance levels. For example, the MS-COCO [127] and NUS-WIDE [128] datasets are widely adopted for evaluating performance under naturally long-tailed distributions and complex label co-occurrence patterns. In medical domains, ChestX-ray8 [129] and CheXpert [130] provide important testbeds for scenarios involving rare pathologies and severe class imbalance. In addition, datasets including DeepFashion [131], Open Images [132], and RSICD [133] further extend evaluation to diverse application domains, enabling the assessment of domain-specific adaptability. However, compared with traditional ML-CIL, the deep learning paradigm exhibits a greater lack of standardized and integrated software tools. Consequently, researchers often rely on customized preprocessing workflows and task-specific data handling strategies, which may further affect the reproducibility of experimental results.

Table X summarizes the aforementioned datasets and software tools.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE X
LIST OF DATASETS AND SOFTWARE TOOLS

No.	Name	URL	Type
1	UCI [114]	https://archive.ics.uci.edu/	Public data repositories
2	Kaggle [115]	https://www.kaggle.com/	
3	OpenML [116]	https://www.openml.org/	
4	MULAN [117]	https://mulan.sourceforge.net/	Integrated software tools
5	MEKA [118]	https://sourceforge.net/projects/meka/	
6	mldr [119]	https://github.com/fcharte/mldr	
7	scikit-multilearn [120]	http://scikit.ml/	
8	MLC Toolbox [121]	https://github.com/KKimura360/MLC_toolbox/	
9	KDIS Lab	http://www.uco.es/kdis/mllresources/	Special research projects
10	Cometa [122]	http://cometa.ujaen.es/	
11	LABIC	http://computer.njnu.edu.cn/Lab/LABIC/LABIC_Software.html	Specific repositories
12	CLUS [123]	https://dtai.cs.kuleuven.be/software/clus/hmcdatasets/	
13	XML [124]	https://manikvarma.org/downloads/XC/XMLRepository.html	
14	MS-COCO [127]	http://mscoco.org/	Representative deep multi-label datasets
15	NUS-WIDE [128]	http://lms.comp.nus.edu.sg/research/NUS-WIDE.htm	
16	ChestX-ray8 [129]	https://www.cc.nih.gov/drd/summers.html	
17	CheXpert [130]	https://stanfordmlgroup.github.io/competitions/chexpert/	
18	DeepFashion [131]	http://mmlab.ie.cuhk.edu.hk/projects/DeepFashion.html	
19	Open Images [132]	https://g.co/dataset/openimages/	
20	RSICD [133]	https://github.com/201528014227051/RSICD_optimal	

VIII. APPLICATIONS AND FUTURE DEVELOPMENT DIRECTIONS

A. Applications

In the previous sections, we have systematically reviewed various algorithms and techniques, which offer strong theoretical support for tackling the ML-CIL problem. Recently, driven by technological advancements and practical needs, ML-CIL algorithms have shown significant potential across various domains, notably advancing applications in industry, medicine, and biology. In industrial scenarios, ML-CIL is widely used for anomaly detection, including classifying surface defects [134], diagnosing electric vehicle malfunctions [135], detecting short circuits in electrolytic refining [136], and enhancing data quality in industrial IoT [137]. In medicine, ML-CIL improves diagnostic accuracy and efficiency, with applications in rhinitis prediction [138], miRNA-related disease diagnosis [139], stroke detection [140], and generating diagnostic reports from chest X-rays [129], [130]. In biology, ML-CIL is applied to identify antimicrobial peptides (AMPs) and their functions [141], predict imbalanced multi-label protein subcellular localization in immunohistochemistry images [142], and predict multiple lysine modification sites [143]. Beyond these fields, ML-CIL is also used in daily-life scenarios such as multi-label sentiment analysis in natural language processing [144], dynamic food ingredient recognition [145], pedestrian attribute recognition [146], and road scene recognition [147].

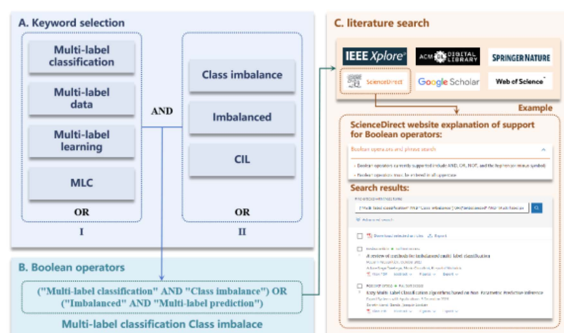

Fig. 6. Schematic of literature search steps.

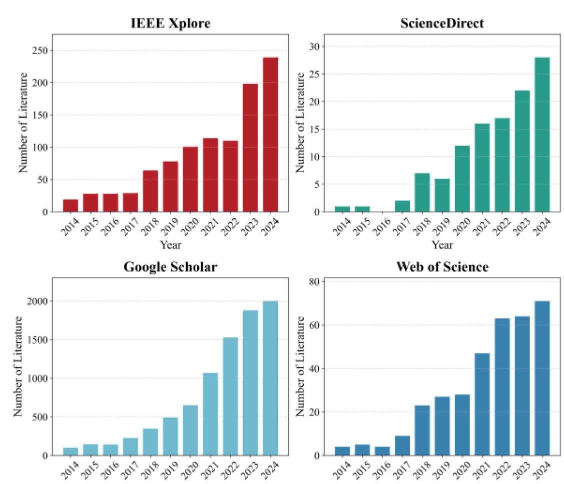

Fig. 7. Trends in ML-CIL publication counts across four main academic databases from 2014 to 2024.

B. Future Development Directions

Current methods have effectively tackled ML-CIL challenges using various technical strategies, but they remain limited and are still evolving. As research advances, certain trends are emerging that may become important directions for the future development of ML-CIL.

1) In traditional machine learning, the development of ML-CIL methods has shifted from treating labels independently to explicitly modeling label correlations. The strategies for capturing these correlations have evolved from random selection to heuristic approaches. Accordingly, future ML-CIL algorithms are expected to explore label distributions more systematically, leverage latent label dependencies more effectively, and integrate correlation modeling more deeply into classifier design.

2) In addition, different ML-CIL technical approaches have shown distinct development trends. Random sampling strategies have largely been abandoned in favor of heuristic oversampling, which shows greater potential than undersampling. Emerging sample synthesis techniques and concepts like local imbalance and reverse nearest neighbor are being explored to better capture prior label distributions, with increased focus on sampling boundaries. Hybrid sampling methods, combining undersampling and oversampling, are also a significant research direction. In cost-sensitive learning, despite its dependency on specific classifiers, researchers are now increasingly introducing label weights into the training process in various ways, rather than simply modifying the loss function. Furthermore, the combination of optimization methods and ensemble learning frameworks may be a future trend in cost-sensitive learning. Currently, threshold moving methods specifically developed for ML-CIL problems are limited in number and form, and remain in an early stage of development. In the future, improvements commonly seen in other strategies are expected to be transferred to this area, and probabilistic or ranking-wise threshold moving approaches for ML-CIL are also likely to emerge.
3) As for other single methods, the migration of new technologies and the adaptation of classifiers will be used to solve ML-CIL problems, which represents a corresponding development trend. In the context of ensemble learning, multi-label datasets typically offer sufficient label and feature complexity to support the training of diverse classifiers, whether in heterogeneous or homogeneous settings. Moreover, integration strategies continue to evolve—from simple ensemble combinations to meta-learning and cross-coupling. According to ensemble learning theory, future advancements will likely focus on increasing the diversity and quality of base classifiers, along with the development of novel ML-CIL ensemble frameworks.
4) In deep learning, solutions to ML-CIL problems—whether general or extreme—are gradually shifting from traditional strategies to feature learning. Feature learning enables automatic extraction of complex data patterns, facilitates more efficient modeling of label correlations through shared layers or deep representations, and allows for adaptive model adjustments, thereby significantly improving generalization performance. Moreover, modeling label dependencies remains a central focus in deep learning–based ML-CIL research.
5) A key future direction is to develop dedicated ML-CIL methods for fine-grained scenarios, including semi-supervised learning, active learning, and continual learning. Taking active learning with cost-sensitive learning as an example, an imbalance-aware mechanism can be introduced into the query function by defining the query score of a sample as a weighted combination of label-wise uncertainties, where the weights are based on the inverse of label frequency or class cost. This design increases the chance of selecting samples that contain minority labels, thereby reducing imbalance during data collection and improving annotation quality. Similar ideas can be extended to other scenarios, such as using imbalance-aware confidence for pseudo-labeling in semi-supervised learning, or applying cost-sensitive replay and regularization in continual learning to better preserve minority label information and achieve more stable model performance.
6) The development of ML-CIL methods tailored to specific application domains is expected to be a key future direction. Unlike general-purpose benchmarks, real-world scenarios often involve domain-specific label semantics, label noise, or dynamic distributions, necessitating more specialized solutions. For example, medical applications prioritize interpretability and uncertainty estimation; industrial settings demand efficient models under data sparsity and latency constraints; while fields such as remote sensing face challenges related to multimodal data and spatiotemporal dependencies. Therefore, future research will increasingly focus on designing customized approaches that integrate domain knowledge, data characteristics, and task-specific requirements into the core of ML-CIL model development.

IX. Concluding Remarks

This paper presents a comprehensive review of multi-label class imbalance learning, systematically covering problem descriptions, formal definitions, representative algorithms, evaluation metrics, datasets, and software tools. Compared with existing reviews [14], this work significantly expands the research scope by analyzing 70 core algorithms and provides the first systematic survey of deep learning-based ML-CIL methods. Furthermore, we propose a novel multi-level taxonomy that organizes existing technical routes into five major categories: sampling, cost-sensitive learning, threshold moving, other single methods, and ensemble learning. These categories are further refined into subcategories to reveal their underlying algorithmic mechanisms. In addition to the aforementioned systematic analysis, we construct a detailed practical guide to assist researchers in selecting appropriate algorithms, evaluation metrics, datasets, and tools. Finally, we summarize the applications of ML-CIL across various domains and discuss future research trends. This review aims to fill an important gap in existing ML-CIL surveys, provide a valuable reference for future studies, and promote the development of next-generation ML-CIL methods.

Appendix

To comprehensively assess the current development and research trends in ML-CIL, we develop a novel Boolean-based literature search strategy in this review. As illustrated in Fig. 6, we first define keywords in Step A and convert them into a unified Boolean expression in Step B. The logic is based on the co-occurrence of terms from both domains (e.g., "("multi-label classification" AND "class imbalance") OR ("imbalanced" AND "multi-label prediction")"). Finally, Step C performs the search across major databases, covering journal articles, conference papers, and books.

Fig. 7 presents the search results across four major academic databases from 2014 to 2024, showing a steady increase in ML-CIL-related publications. Despite minor overlaps among databases, the period from 2019 to 2020 marks a clear turning point, after which the number of papers grows rapidly and reaches a peak in 2024. This trend reflects increasing academic interest and highlights ML-CIL as a promising research direction.

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